

## HUSSAR AGDM, AB, Canada

WMO: 713440

Lat: 51.1911N

Lon: 112.5027W

Elev: 971

StdP: 90.19

Time Zone: -7.00 (NAM)

Period: 02-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | -28.2      | -25.4 | -31.8                         | 0.2 | -28.1 | -28.5 | 0.3 | -25.2 | 14.1                  | -4.0 | 12.6 | -5.6 | 4.4                   | 320  | 0.746 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 12.7                   | 28.7            | 16.3 | 26.7 | 15.6 | 24.8 | 15.0 | 18.4                | 25.0 | 17.2 | 23.9 | 16.2 | 22.8 | 4.2                  | 210  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 16.0                            | 12.8 | 22.2 | 14.7 | 11.7 | 20.3 | 13.5 | 10.8 | 19.0 | 56.4          | 25.1 | 52.2 | 24.0 | 48.9 | 22.9 | 23.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 12.8          | 11.1 | 9.8 | DB                         | -31.9 | 32.1               | 2.6 | 2.1                                                | -33.7 | 33.6       | -35.2 | 34.8       | -36.6 | 35.9       | -38.5 | 37.4 | (4) |
| (5)               |      |     | WB                         | -32.0 | 21.5               | 2.5 | 1.4                                                | -33.7 | 22.4       | -35.2 | 23.2       | -36.6 | 24.0       | -38.4 | 25.0 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 3.9         | -7.8  | -8.9  | -2.9  | 3.9   | 10.1  | 13.9  | 17.0  | 15.9  | 11.4  | 4.8   | -3.0  | -7.9  | (6)  |  |
|                                                                        | DBStd       | 11.17       | 8.66  | 8.13  | 7.81  | 5.34  | 4.70  | 3.22  | 3.13  | 3.60  | 4.96  | 5.56  | 7.20  | 7.78  | (7)  |  |
|                                                                        | HDD10.0     | 2900        | 553   | 529   | 402   | 190   | 57    | 4     | 0     | 3     | 41    | 176   | 389   | 556   | (8)  |  |
|                                                                        | HDD18.3     | 5299        | 812   | 763   | 659   | 433   | 256   | 137   | 64    | 91    | 212   | 420   | 639   | 814   | (9)  |  |
|                                                                        | CDD10.0     | 692         | 0     | 0     | 1     | 7     | 61    | 122   | 216   | 187   | 83    | 14    | 1     | 0     | (10) |  |
|                                                                        | CDD18.3     | 49          | 0     | 0     | 0     | 0     | 2     | 4     | 22    | 16    | 5     | 0     | 0     | 0     | (11) |  |
|                                                                        | CDH23.3     | 748         | 0     | 0     | 0     | 2     | 41    | 67    | 296   | 248   | 89    | 3     | 0     | 0     | (12) |  |
|                                                                        | CDH26.7     | 178         | 0     | 0     | 0     | 0     | 5     | 8     | 82    | 63    | 20    | 0     | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 5.0         | 5.7   | 5.2   | 5.2   | 5.3   | 5.4   | 5.0   | 4.1   | 4.1   | 4.7   | 5.0   | 5.3   | 5.2   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 384         | 14    | 10    | 16    | 26    | 49    | 77    | 65    | 54    | 32    | 16    | 14    | 12    | (15) |  |
|                                                                        | PrecMax     | 490         | 34    | 24    | 54    | 60    | 93    | 152   | 151   | 108   | 82    | 35    | 48    | 29    | (16) |  |
|                                                                        | PrecMin     | 229         | 1     | 1     | 3     | 3     | 6     | 33    | 24    | 3     | 5     | 5     | 2     | 1     | (17) |  |
|                                                                        | PrecStd     | 65          | 8     | 6     | 10    | 14    | 24    | 30    | 30    | 28    | 19    | 9     | 11    | 8     | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 8.2   | 8.3   | 15.2  | 21.7  | 27.1  | 27.7  | 31.8  | 30.9  | 29.2  | 23.3  | 14.9  | 8.9   | (19) |  |
|                                                                        |             | MCWB        | 3.0   | 2.8   | 7.2   | 9.9   | 12.5  | 16.5  | 17.6  | 15.3  | 14.4  | 11.3  | 7.7   | 3.5   | (20) |  |
|                                                                        | 2%          | DB          | 5.5   | 5.2   | 11.3  | 18.5  | 24.2  | 25.2  | 29.0  | 28.3  | 26.2  | 18.6  | 11.0  | 6.1   | (21) |  |
|                                                                        |             | MCWB        | 1.8   | 1.5   | 4.9   | 8.3   | 11.8  | 14.9  | 17.7  | 15.6  | 13.5  | 9.5   | 5.2   | 2.4   | (22) |  |
|                                                                        | 5%          | DB          | 4.3   | 3.2   | 8.6   | 15.3  | 21.6  | 23.0  | 26.5  | 26.2  | 23.1  | 15.5  | 8.1   | 4.2   | (23) |  |
|                                                                        |             | MCWB        | 1.3   | 0.2   | 3.6   | 6.5   | 10.6  | 13.9  | 17.3  | 15.4  | 12.5  | 8.0   | 3.7   | 1.0   | (24) |  |
|                                                                        | 10%         | DB          | 2.9   | 1.3   | 6.0   | 12.6  | 19.0  | 21.1  | 24.4  | 23.9  | 20.1  | 12.7  | 5.7   | 2.3   | (25) |  |
|                                                                        |             | MCWB        | 0.4   | -1.1  | 2.2   | 5.1   | 9.4   | 13.1  | 16.5  | 15.0  | 11.3  | 6.7   | 2.0   | -0.5  | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 3.6   | 3.5   | 7.8   | 10.4  | 13.7  | 18.4  | 21.3  | 18.7  | 15.4  | 11.8  | 7.6   | 4.2   | (27) |  |
|                                                                        |             | MCDB        | 7.0   | 7.1   | 14.0  | 20.9  | 24.5  | 25.0  | 27.7  | 24.8  | 25.4  | 21.8  | 14.5  | 7.5   | (28) |  |
|                                                                        | 2%          | WB          | 2.2   | 1.7   | 5.5   | 8.6   | 12.3  | 16.4  | 19.3  | 17.4  | 14.1  | 10.1  | 5.5   | 2.6   | (29) |  |
|                                                                        |             | MCDB        | 5.0   | 4.9   | 10.5  | 16.9  | 21.7  | 22.7  | 25.8  | 24.5  | 23.6  | 16.8  | 10.2  | 5.9   | (30) |  |
|                                                                        | 5%          | WB          | 1.4   | 0.3   | 3.8   | 7.1   | 11.3  | 14.9  | 18.0  | 16.4  | 12.9  | 8.5   | 4.0   | 1.0   | (31) |  |
|                                                                        |             | MCDB        | 4.1   | 2.9   | 7.9   | 14.2  | 19.8  | 20.9  | 24.6  | 23.7  | 21.9  | 15.0  | 7.6   | 4.1   | (32) |  |
|                                                                        | 10%         | WB          | 0.3   | -1.0  | 2.3   | 5.7   | 10.2  | 13.8  | 16.9  | 15.4  | 11.7  | 7.0   | 2.4   | -0.4  | (33) |  |
|                                                                        |             | MCDB        | 2.9   | 1.0   | 5.7   | 11.6  | 17.6  | 19.7  | 23.3  | 22.4  | 19.3  | 12.3  | 5.3   | 2.2   | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 8.6   | 8.6   | 9.1   | 11.1  | 12.0  | 11.1  | 12.7  | 12.7  | 12.1  | 10.8  | 8.6   | 8.1   | (35) |  |
|                                                                        |             | MCDBR       | 8.8   | 10.0  | 12.5  | 16.0  | 15.5  | 14.3  | 15.9  | 16.6  | 16.6  | 14.6  | 11.5  | 8.9   | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 6.4   | 7.3   | 7.6   | 8.0   | 6.8   | 6.7   | 8.0   | 7.0   | 7.1   | 7.6   | 7.2   | 6.4   | (37) |  |
|                                                                        |             | MCWBR       | 8.8   | 9.7   | 12.0  | 14.8  | 13.8  | 12.6  | 14.2  | 14.3  | 15.3  | 14.1  | 11.4  | 9.1   | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.260 | 0.268 | 0.280 | 0.334 | 0.359 | 0.351 | 0.349 | 0.366 | 0.320 | 0.279 | 0.250 | 0.239 | (40) |  |
|                                                                        |             | taud        | 2.381 | 2.435 | 2.479 | 2.335 | 2.311 | 2.382 | 2.416 | 2.345 | 2.449 | 2.545 | 2.498 | 2.407 | (41) |  |
|                                                                        | Edn at Noon | Edn at Noon | 783   | 881   | 934   | 906   | 894   | 903   | 899   | 863   | 873   | 860   | 803   | 758   | (42) |  |
|                                                                        |             | Edh at Noon | 59    | 74    | 86    | 113   | 121   | 114   | 109   | 111   | 89    | 66    | 52    | 49    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 1.14  | 2.15  | 3.51  | 4.77  | 5.75  | 6.20  | 6.66  | 5.37  | 3.79  | 2.30  | 1.24  | 0.88  | (44) |  |
|                                                                        | RadStd      | RadStd      | 0.09  | 0.18  | 0.29  | 0.36  | 0.39  | 0.38  | 0.33  | 0.37  | 0.38  | 0.27  | 0.10  | 0.08  | (45) |  |

## Historical Trends

|                         |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| Station Only            |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
|                         |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| Regional (81 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page